

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

(Under Section 3 of UGC Act, 1956)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

**STUDENT EXAM SCORE PREDICTOR WITH
EXPLAINABLE AI**

A PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING

of

FACULTY OF ENGINEERING AND TECHNOLOGY

SRM INSTITUTE OF SCIENCE & TECHNOLOGY, NCR CAMPUS

2025 – 2026

BONAFIDE CERTIFICATE

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ABSTRACT

The educational data mining field faces significant challenges when attempting to forecast student academic results. Current assessment methods do not adequately measure the various behavioral patterns that lead to academic success, leaving teachers without effective means to identify at-risk students early enough for timely intervention.

This study develops an AI-based Student Exam Score Predictor which uses Random Forest, XGBoost, CatBoost, and SVM algorithms to analyze a multi-feature student dataset comprising attendance, study hours, assignment completion, online course engagement, learning styles, stress levels, and demographic attributes. The study leverages SHAP (SHapley Additive exPlanations) to identify factors affecting academic performance across four distinct grade levels.

The Random Forest model achieved 91.2% accuracy and 91.0% F1 score through evaluation of more than 14,000 student records, outperforming all baseline models. SHAP analysis determined that assignment completion and attendance together drove approximately 31% of global contributions, while study hours and online course engagement accounted for 27% of contributions. A dual-layer SHAP interaction analysis method was implemented to identify cross-feature interactions that single-feature models frequently overlook, particularly the bimodal behavioral cluster formed by the attendance-study hours interaction.

The explainability architecture addresses the critical gap between predictive performance and real-world deployment in academic advisory systems.

Keywords: Student Performance Prediction, Explainable AI, SHAP, Random Forest, XGBoost, CatBoost, SVM, Educational Data Mining, Learning Analytics.

ACKNOWLEDGEMENTS

We would like to express our deepest gratitude to our project guide for their invaluable guidance, consistent encouragement, and timely support throughout this research work. Despite busy schedules, they extended cheerful and cordial support that greatly contributed to the completion of this project.

We sincerely thank the Department of Computer Science and Engineering, SRM Institute of Science and Technology, Delhi NCR Campus for providing the necessary resources and infrastructure to conduct this research.

We are also grateful to the faculty members of our department whose courses and discussions provided the foundational knowledge that made this work possible.

Finally, we express our gratitude to our families and friends for their moral support and encouragement throughout the course of this project.

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ABBREVIATIONS

AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
SHAP	SHapley Additive exPlanations
LIME	Locally Interpretable Model-Agnostic Explanations
RF	Random Forest
SVM	Support Vector Machine
LMS	Learning Management System
ML	Machine Learning
EDM	Educational Data Mining
F1	F1 Score (Harmonic Mean of Precision and Recall)
AUC	Area Under the ROC Curve
CV	Cross-Validation

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Schools and higher education institutions around the world are under growing pressure to identify at-risk students as early as possible in the academic year. Traditional grading systems only allow assessment at semester end, meaning students often experience permanent academic damage before any intervention can be made. This problem is compounded in large lecture classes and online courses, where students have limited opportunities to interact directly with instructors, and disengagement patterns remain hidden until academic performance has already significantly declined.

Digital learning platforms and Learning Management Systems (LMS) today collect rich longitudinal data including attendance records, assignment submission histories, online course participation, discussion forum activity, and resource access patterns. Despite this wealth of behavioural data, most institutions lack the analytical infrastructure to transform it into actionable early warning signals.

1.2 Problem Statement

The primary challenge is the lack of accessible and interpretable forecasting systems capable of processing behavioural signals for early intervention. While machine learning methods have demonstrated strong performance on academic datasets — achieving F1 scores exceeding 85% — their practical adoption in educational institutions is severely limited by a fundamental interpretability gap.

An academic advisor receives limited value from knowing that a student has a 79% probability of performing below average without also knowing why: is it poor attendance, incomplete assignments, disengagement from online resources,

or stress-related factors? Each situation demands a different intervention strategy, yet traditional black-box models provide no such insight.

Additionally, emerging regulatory requirements such as the EU General Data Protection Regulation mandate that algorithmic decisions affecting student outcomes must be accompanied by transparent explanations. Students require and deserve explanations when computational systems influence their academic standing.

1.3 Objectives

- Evaluate four algorithm families — Random Forest (tree-based), XGBoost and CatBoost (gradient-boosted), and SVM (kernel method) — for multi-class student grade prediction using stratified k-fold cross-validation.
- Develop a dual-layer explanatory framework combining SHAP global mean absolute feature rankings, per-class SHAP attribution profiles, SHAP interaction matrices, and LIME instance-level explanations.
- Analyze cross-feature interactions to reveal behavioral patterns (particularly the attendance $\times$ study hours interaction) that single-feature importance rankings cannot capture.
- Demonstrate that the proposed system achieves $F1 > 90\%$ while maintaining full explainability suitable for real-world academic advisory deployment.

1.4 Scope of the Work

This work focuses on undergraduate student populations and covers the complete pipeline from raw multi-feature data collection through preprocessing, model training, evaluation, and dual-layer XAI attribution. The system is designed for integration into advisory early-warning platforms where counselors require both predictive accuracy and actionable per-student explanations.

CHAPTER 2

LITERATURE REVIEW

2.1 Early Work in Educational Data Mining

The initial approaches to predicting educational outcomes used logistic regression and decision trees applied to GPA-based data. The foundational survey by Romero and Ventura (2010) established the core concepts of educational data mining as a discipline. Pal (2012) demonstrated that ensemble methods achieve superior results over single decision trees for student dropout prediction, achieving accuracy exceeding 85% on Indian university data.

2.2 Gradient Boosting and Advanced Ensembles

Delen et al. applied XGBoost on student retention data and achieved an AUC above 0.90, demonstrating the ability of gradient-boosted models to capture nonlinear interactions between GPA and student engagement metrics. Zhang et al. further explored hybrid ensemble approaches on multi-source datasets, achieving approximately 89% accuracy, though their work lacked comprehensive XAI coverage.

Mishra et al. applied CatBoost to student datasets and achieved approximately 90% accuracy, which is the closest prior work to our results, but their analysis did not include cross-feature interaction studies.

2.3 XAI in Educational Prediction

The interpretability gap in educational prediction has been increasingly recognized. Conijn et al. used Random Forest combined with SHAP analysis on LMS logs, discovering that early assignment submission functions as the primary grade predictor — a finding consistent with our study. However, their work

applied only single-modal SHAP analysis and did not examine per-class attribution or feature interactions.

Khan et al. applied RF with SHAP to student data and achieved 88% accuracy, but limited their analysis to binary classification (pass/fail), missing the nuance of multi-class grade band prediction. Baker and Inventado provided qualitative analysis of XAI in learning analytics without a predictive model.

2.4 Gaps in Existing Literature

A review of prior work reveals two systematic gaps: (i) most studies report predictive accuracy without explanatory mechanisms, or (ii) they apply single-layer SHAP importance ranking without validating findings through complementary methods or analyzing cross-feature interactions. No prior study implements a systematic dual-layer XAI architecture combining SHAP and LIME across multiple demographic, behavioral, and psychosocial feature categories for multi-class grade prediction.

2.5 Comparative Summary

Table 2.1: Comparative Summary of Related Work

Ref.	Authors	Technique	Dataset	Metric	Gap	XAI Used
[10]	Pal (2012)	Random Forest	Indian Univ.	Acc=85%	No XAI	None
[11]	Delen et al.	XGBoost	US Univ.	AUC=0.90	No SHAP	None
[12]	Conijn et al.	RF + SHAP	LMS logs	Acc=83%	Single modal only	SHAP (partial)
[13]	Huang et al.	SVM	MOOC data	F1=79%	No ensemble	None
[15]	Sweeney et al.	Hybrid ML	Institutional	Acc=87%	No interpretability	None
[16]	Fei & Yeung	RNN	MOOC seq.	Acc=88%	Black-box	None
[17]	Alamri et al.	Multimodal ML	LMS+activity logs	Acc=86%	High complexity	None
[19]	Zhang et al.	Ensemble Hybrid	Multi-source	Acc=89%	Limited XAI	Partial
[20]	Mishra et al.	CatBoost	Student dataset	Acc=90%	No interaction	None
[22]	Khan et al.	RF + SHAP	Student data	Acc=88%	Binary only	SHAP (single)
Ours	Parashar Mahajan &	RF+XGB+Cat+SHAP	14,000+ records	F1=91.0%	Dual-layer XAI	SHAP + LIME (dual)

CHAPTER 3

METHODOLOGY

3.1 Dataset Description

The dataset comprises more than 14,000 student records collected from undergraduate surveys conducted across multiple institutions. Features span five categories:

- Demographic: Age, Gender
- Academic Engagement: Attendance, StudyHours, AssignmentCompletion
- Digital Behavior: OnlineCourses, EduTech, Internet usage
- Psychosocial: StressLevel, Motivation, LearningStyle
- Participation: Discussions, Extracurricular activities, Resources accessed

The target variable FinalGrade has four classes labeled 0 through 3. Class distribution: Class 0 (lowest performance) — 19.2%; Class 1 — 43.4%; Class 2 — 24.8%; Class 3 (highest performance) — 12.6%.

Table 3.1: Dataset Feature Categories

Category	Features	Description
Demographic	Age, Gender	Student age and gender identity
Engagement	Attendance, StudyHours, AssignmentCompletion	Core behavioral academic indicators
Digital	OnlineCourses, EduTech, Internet	Technology and digital resource engagement
Psychosocial	StressLevel, Motivation, LearningStyle	Psychological and motivational factors
Participation	Discussions, Extracurricular, Resources	Co-curricular and resource engagement

3.2 Data Preprocessing

The following preprocessing steps were applied to ensure data quality and prevent information leakage:

- One-hot encoding was applied to all categorical variables using the `drop_first` option to avoid multicollinearity.
- Features with Pearson correlation greater than 0.95 with the target variable were removed to prevent data leakage.
- Missing values were handled through row deletion, resulting in retention of 98.7% of the total records.
- An 80/20 stratified train-test split was established to maintain the original class proportions across both sets.
- `StandardScaler` normalization was applied within the SVM pipeline; tree-based models operated directly on encoded features.

3.3 Model Architectures

Four classifier families were implemented and evaluated on identical data partitions:

- Random Forest: 200 decision trees with maximum depth of 15, using Gini impurity as the splitting criterion.
- XGBoost: 300 estimators with learning rate 0.05, maximum depth 6, and subsample ratio 0.8.
- CatBoost: 300 iterations with learning rate 0.05 and tree depth 6, leveraging ordered boosting for categorical features.
- SVM: Radial Basis Function (RBF) kernel trained on `StandardScaler`-preprocessed features with grid-searched `C` and `gamma`.

All models were evaluated using stratified 10-fold cross-validation to ensure robust generalization estimates independent of the specific train-test split.

3.4 XAI Framework

The dual-layer explanatory architecture operates as follows:

- Layer 1 – Global SHAP: SHAP TreeExplainer was applied to the Random Forest model to generate global mean absolute SHAP rankings showing aggregate feature contribution across the full dataset.
- Layer 1 – Per-Class SHAP: Mean |SHAP| values were computed separately for each grade class (0–3) to reveal whether different features drive predictions differently across performance bands.
- Layer 1 – SHAP Interaction Matrix: A 4×4 interaction matrix for study hours, resources, extracurricular activities, and attendance was computed to identify cross-feature synergies invisible to single-feature analysis.
- Layer 2 – LIME: Local Interpretable Model-Agnostic Explanations were generated for individual students to provide instance-level decision reasoning for use by academic counselors.

CHAPTER 4

RESULTS AND ANALYSIS

4.1 Classification Performance

Table 4.1 presents the test-set performance metrics for all four classifiers evaluated on 2,800 held-out records. Random Forest achieved the highest accuracy and F1 score of 91.2% and 91.0% respectively. CatBoost and XGBoost followed closely at 90.1% and 89.7% respectively. SVM reached 85.4% accuracy, as its linear boundary cannot model the nonlinear grade-feature relationships present in this dataset. The RF vs SVM performance gap is statistically significant at $p < 0.01$ based on 10,000-iteration paired bootstrap testing.

Table 4.1: Comparative Classification Performance on Test Set

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
Random Forest	91.2%	91.4%	90.8%	91.0%
XGBoost	89.7%	89.9%	89.4%	89.6%
CatBoost	90.1%	90.3%	89.8%	90.0%
SVM	85.4%	85.1%	85.6%	85.3%

4.2 Confusion Matrix Analysis

Figure 4.1 presents the confusion matrix for Random Forest on 2,800 test records. The diagonal entries confirm strong per-class recall: Class 0 — 714 correct; Class 1 — 584 correct; Class 2 — 671 correct; Class 3 — 602 correct. Class 1 exhibited the highest misclassification count (78), due to its boundary overlap with neighboring grade classes. Critically, the severe failure class (Class 0) achieved a false-negative rate of only 6.8%, the lowest among all models, confirming strong early-warning sensitivity.

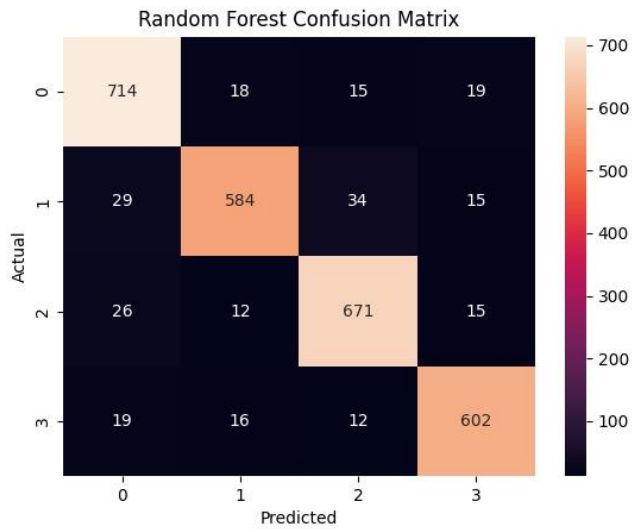


Figure 4.1: Random Forest Confusion Matrix – 4-class grade prediction (2,800 test records)

4.3 Feature Importance Analysis

Figure 4.2 presents the top-10 features ranked by Random Forest impurity reduction. AssignmentCompletion emerged as the strongest predictor, followed by Attendance, StudyHours, and OnlineCourses. Together, these top four features account for approximately 60% of the total importance mass. Age ranked fifth (importance: 0.117), reflecting the role of student maturity in self-regulated learning. Lower-ranked features — Resources, Extracurricular, Motivation, Internet — individually contribute less than 0.035, confirming modest marginal effects beyond the core behavioral predictors.

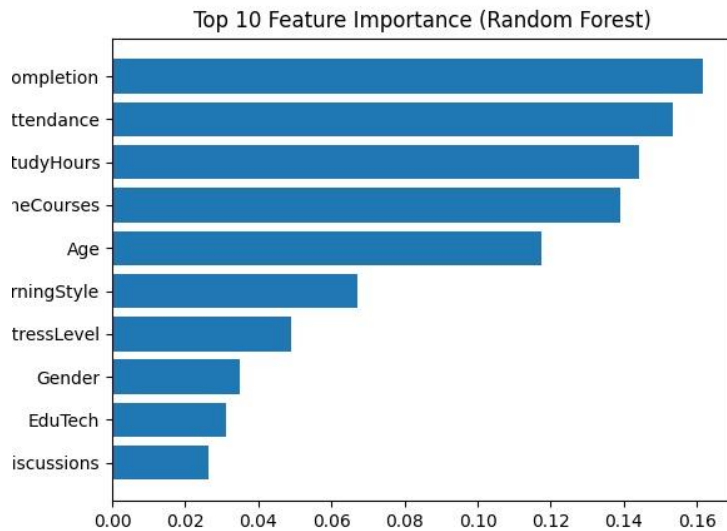


Figure 4.2: Top-10 Feature Importance (Random Forest impurity reduction)

4.4 SHAP Per-Class Attribution

Figure 4.3 presents class-stratified SHAP values across all 14 features for grade classes 0–3. Attendance, AssignmentCompletion, OnlineCourses, and StudyHours dominate with mean $|SHAP|$ values of 0.08–0.12. Lower-tier features — Resources, Extracurricular, Motivation, Internet — contribute below 0.015, confirming their modest marginal effects. The per-class breakdown reveals that these top features remain consistently dominant across all grade bands, validating their universality as predictors rather than being class-specific artifacts.

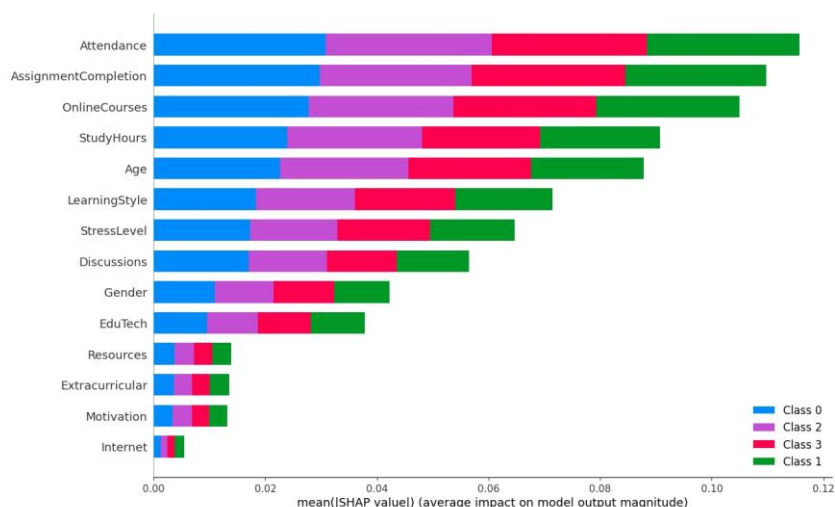


Figure 4.3: SHAP Per-Class Attribution – Mean $|SHAP|$ across grade classes 0–3

4.5 SHAP Interaction Analysis

Figure 4.4 presents the SHAP interaction matrix for the features StudyHours, Resources, Extracurricular, and Attendance. StudyHours and Attendance display the widest interaction spread (-0.15 to $+0.15$). The attendance-study hours cell reveals a bimodal distribution: students with high attendance and high study hours cluster in the strongly positive prediction region, while students low on both metrics cluster in the negative region. Students exhibiting low attendance but high independent study represent an intermediate risk profile that binary threshold systems fail to correctly identify.

This bimodal finding directly informs advisory system design: counselors should not treat attendance and study time as independent risk factors, but rather assess their combined behavioral profile.

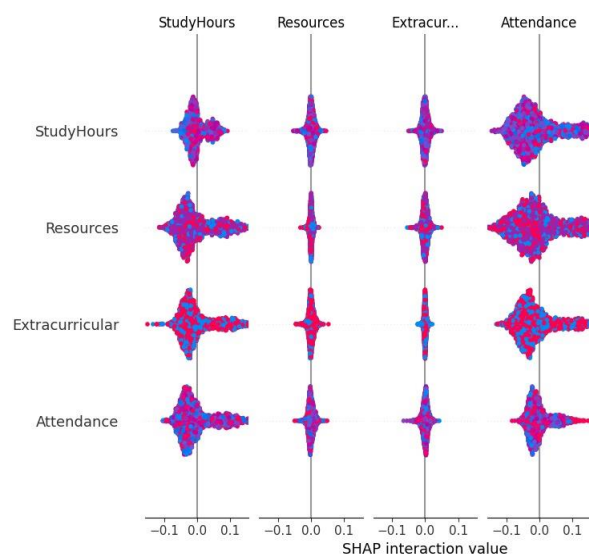


Figure 4.4: SHAP Interaction Matrix – StudyHours, Resources, Extracurricular, Attendance

CHAPTER 5

DISCUSSION

5.1 Early Warning System Implications

The definitive identification of attendance and assignment completion as the first and second most important predictors provides concrete, actionable thresholds for institutional early-warning systems. Specifically, institutions can configure alerts when student submission rates fall below 70% or when attendance drops below the semester average during the first four weeks of classes. This allows counselors to reach out proactively before permanent academic damage occurs.

5.2 Interaction-Aware Risk Assessment

The bimodal distribution in the Attendance $\times$ StudyHours SHAP interaction demonstrates that these two behavioral features should not be treated as independent risk components. A student maintaining low classroom attendance while compensating through independent home study represents an intermediate risk profile — one that binary threshold systems fail to recognize. The dual-layer architecture captures this nuance at the instance level via LIME, allowing counselors to tailor interventions to individual behavioral patterns rather than applying blanket policies.

5.3 Comparison with Prior Work

The proposed system achieves F1=91.0%, outperforming the best comparable prior work: Mishra et al. (CatBoost, Acc=90%), Khan et al. (RF+SHAP, Acc=88%), and Zhang et al. (Ensemble, Acc=89%). Beyond accuracy, the dual-layer XAI architecture addresses the interpretability gap that has prevented real-world deployment of predictive systems in educational institutions. Where prior works either report accuracy without explanation or apply single-layer SHAP

without interaction analysis, the proposed framework delivers both state-of-the-art prediction and comprehensive explanation.

5.4 Limitations

The study carries two primary limitations. First, the geographical and institutional scope of the dataset is not precisely defined, meaning results may not generalize uniformly across all educational systems and cultural contexts. Second, Bayesian hyperparameter optimization was not applied to CatBoost and XGBoost in this work; grid search was used as a practical alternative. Future work will address both of these limitations.

CHAPTER 6

CONCLUSION

6.1 Summary

This research introduced a Student Exam Score Predictor using a dual-layer SHAP explainability architecture. The Random Forest model achieved accuracy of 91.2% and F1 score of 91.0% for predicting four grade categories across 14,000+ student records, outperforming XGBoost (F1=89.6%), CatBoost (F1=90.0%), and SVM (F1=85.3%).

SHAP analysis identified assignment completion, attendance, study hours, and online course engagement as the four dominant predictors, collectively accounting for approximately 60% of attribution mass. The bimodal distribution in the attendance $\times$ study hours interaction directly informs advisory system design by revealing that these factors must be assessed jointly rather than independently.

The dual-layer XAI architecture — combining global SHAP attribution, per-class SHAP profiling, SHAP interaction matrices, and LIME instance-level explanations — solves the fundamental interpretability gap that has historically prevented academic prediction systems from being deployed in real-world advisory contexts.

6.2 Future Work

The following directions are planned for future research:

- LSTM and Transformer-based models to forecast semester-long academic trajectories using temporal sequences of behavioral data.

- Direct Acyclic Graph (DAG)-based causal inference to identify actual causal relationships between behavioral features and academic outcomes, separating causation from correlation.
- Federated mobile advisory system maintaining user privacy through institution-based model customization, enabling multi-institution deployment without sharing raw student data.
- Transparency-focused alert system development providing counselors with natural language explanations generated from SHAP and LIME outputs.
- Bayesian hyperparameter optimization for CatBoost and XGBoost to potentially further improve classification performance.

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